

# Implementation of Relational Database Based on Case Study

## Case Study: Student Database in SQL Server

### 1. Create Database and Table

```
CREATE DATABASE Student;
USE Student;

CREATE TABLE Student (
    StudentID INT PRIMARY KEY IDENTITY(1,1),
    Name NVARCHAR(100),
    FatherName NVARCHAR(100),
    FatherOccupation NVARCHAR(100),
    District NVARCHAR(50),
    Department NVARCHAR(50),
    CGPA DECIMAL(3,2)
);
```

### 2. Insert Data

```
INSERT INTO Student (Name, FatherName, FatherOccupation, District, Department, CGPA)
VALUES
('Rahim Uddin', 'Karim Uddin', 'Teacher', 'Dhaka', 'CSE', 3.75),
('Sadia Akter', 'Shafiq Akter', 'Doctor', 'Chittagong', 'EEE', 3.60),
('Mizan Rahman', 'Jamal Rahman', 'Businessman', 'Dhaka', 'BBA', 3.20),
('Nusrat Jahan', 'Anis Jahan', 'Engineer', 'Sylhet', 'CSE', 3.90),
('Tariq Islam', 'Nurul Islam', 'Farmer', 'Rajshahi', 'CSE', 2.95);
```

### 3. Update Data

```
UPDATE Student
SET CGPA = 3.85
WHERE StudentID = 1;
```

### 4. Delete Data

```
DELETE FROM Student
WHERE StudentID = 5;
```

### 5. Distinct Example

```
SELECT DISTINCT District
FROM Student;
```

### 6. UNION vs UNION ALL

```
-- UNION (removes duplicates)
SELECT Name, Department FROM Student WHERE Department = 'CSE'
UNION
SELECT Name, Department FROM Student WHERE Department = 'EEE';
```

```
-- UNION ALL (keeps duplicates)
SELECT Name, Department FROM Student WHERE Department = 'CSE'
UNION ALL
SELECT Name, Department FROM Student WHERE Department = 'EEE';
```

## 7. CASE Command Example

```
SELECT Name, CGPA,
       CASE
           WHEN CGPA >= 3.75 THEN 'A'
           WHEN CGPA >= 3.50 THEN 'B+'
           WHEN CGPA >= 3.00 THEN 'B'
           ELSE 'C'
       END AS Grade
FROM Student;
```

## 8. Benefits & Conclusion

- Centralized student data management. - Reduces redundancy and ensures data consistency. - Easy to query by department, district, or CGPA. - Supports analysis & reporting (e.g., top-performing students by grade).